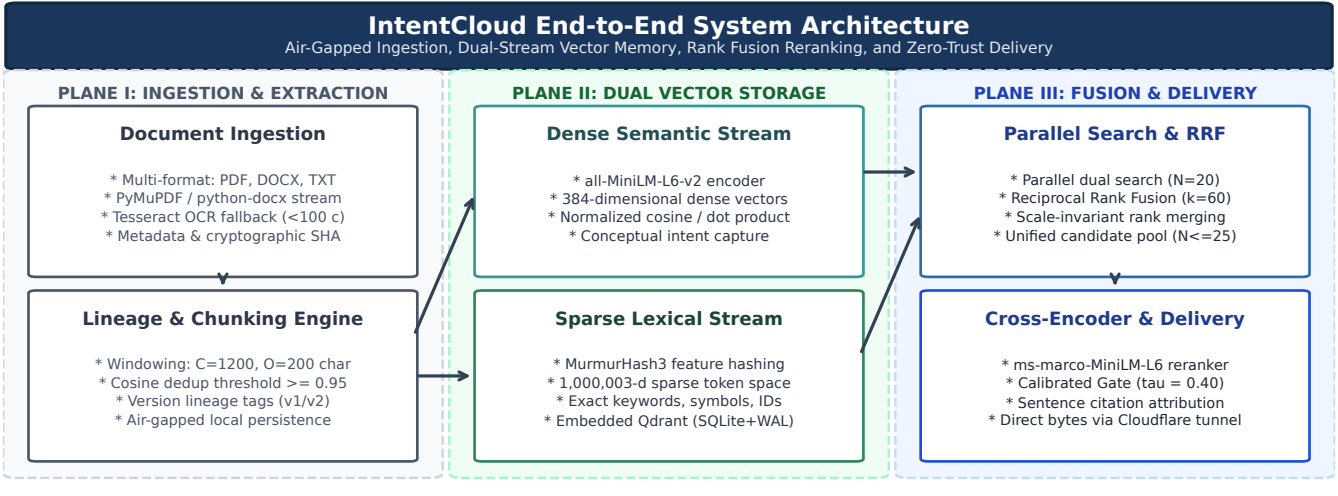
IntentCloud End-to-End System Architecture
Air-Gapped Ingestion, Dual-Stream Vector Memory, Rank Fusion Reranking, and Zero-Trust Delivery
PLANE I: INGESTION & EXTRACTION
Document Ingestion
* Multi-format: PDF, DOCX, TXT
* PyMuPDF / python-docx stream
* Tesseract OCR fallback (<100 c)
* Metadata & cryptographic SHA
Lineage & Chunking Engine
* Windowing: C=1200, O=200 char
* Cosine dedup threshold >= 0.95
* Version lineage tags (v1/v2)
* Air-gapped local persistence
PLANE II: DUAL VECTOR STORAGE
Dense Semantic Stream
* all-MiniLM-L6-v2 encoder
* 384-dimensional dense vectors
* Normalized cosine / dot product
* Conceptual intent capture
Sparse Lexical Stream
* MurmurHash3 feature hashing
* 1,000,003-d sparse token space
* Exact keywords, symbols, IDs
* Embedded Qdrant (SQLite+WAL)
PLANE III: FUSION & DELIVERY
Parallel Search & RRF
* Parallel dual search (N=20)
* Reciprocal Rank Fusion (k=60)
* Scale-invariant rank merging
* Unified candidate pool (N<=25)
Cross-Encoder & Delivery
* ms-marco-MiniLM-L6 reranker
* Calibrated Gate (tau = 0.40)
* Sentence citation attribution
* Direct bytes via Cloudflare tunnel